FastSpeech 2-Based Text-to-Speech Synthesis

English (Svarah) and Telugu (IndicTTS Telugu): technical report

GitHub: https://github.com/LightningHat/FastSpeech-2-Based-Text-to-Speech-Synthesis

Summary and honest status

Two separate FastSpeech 2 models were trained, one per language, starting from the reference implementation named in the assignment. This report describes what was built, how it was trained, and what the outputs actually look and sound like. It deliberately reports failures: where the generated speech is not intelligible, this is stated rather than hidden.

Status of the output quality. The English model, after 35 epochs of training, produces audio that is dominated by a low hum with a faint, hazy voice in the background that is mostly unintelligible. A likely explanation is under-training (the run performed roughly 4,000 optimizer updates, far fewer than the hundreds of thousands normally used for FastSpeech 2) combined with a weak mel-to-waveform stage. This explanation is a hypothesis; the diagnostic tests that would confirm it are listed in Section 6.

1. Implementation approach

1.1 Architecture components used from the reference implementation

The model follows FastSpeech 2 (Ren et al., 2021) as implemented in the reference repository. The components used are:

Component	Role
Phoneme/character encoder	Embedding plus a stack of feed-forward Transformer (FFT) blocks, each with multi-head self-attention and a convolutional feed-forward layer.
Variance adaptor	Three predictors for duration, pitch and energy. Pitch and energy are quantized into bins (stored as model buffers) and embedded, then added to the hidden sequence. Duration drives the length regulator.
Length regulator	Expands each phoneme representation by its (predicted at inference, ground-truth during training) duration in mel frames.
Mel decoder + PostNet	FFT-block decoder produces the mel spectrogram; a convolutional PostNet refines it. Both outputs are trained with L1 loss.
Loss (FastSpeech2Loss)	Mel L1 + PostNet L1 + pitch + energy + log-duration MSE, combined into one total loss.

1.2 Modifications made

Change	Where	Reason
Attention mask value changed from -1e9 to the minimum value of the tensor dtype	core/model.py	-1e9 cannot be represented in float16; under mixed-precision training it raised a RuntimeError (overflow). Applies to both languages.
Inference duration clamp min=1, with rounding applied after the control factor	core/model.py	Prevents a phoneme from receiving 0 frames at inference, which would silently delete it.
Predictor output biases initialised: duration 1.75, pitch = dataset pitch mean, energy = dataset energy mean	train loop	Starts the predictors in a valid range so early training is stable.
Mixed-precision (AMP) training with GradScaler, AdamW, OneCycle LR	train loop	Faster training on a single GPU.
Pitch and energy bin buffers restored from the checkpoint at load time	inference.py	Bins depend on dataset statistics; they must match the trained model, not the defaults.
Pitch from YIN with linear interpolation across unvoiced frames; energy as frame L2 norm of the STFT magnitude	core/audio.py	Fast frame-level F0 and energy targets aligned to the mel frames.
Waveform generation: Griffin-Lim (64 iterations) from the mel via pseudo-inverse of the mel basis	core/audio.py	Reference mel-to-audio path. A pretrained HiFi-GAN vocoder was also tried (see 1.3).

1.3 Vocoder

Both Griffin-Lim (64 iterations) and a pretrained HiFi-GAN universal neural vocoder (jaketae/hifigan-lj-v1) were implemented and evaluated. The submitted test audios were generated using the HiFi-GAN vocoder. While Griffin-Lim produces metallic phase smear, HiFi-GAN provides significantly smoother reconstruction, though both are limited by the acoustic over-smoothing of the 35-epoch mel predictions.

1.4 Language-specific modifications (Telugu)

The Telugu text frontend utilizes Indic NLP Library tokenizer and Morfessor phoneme decomposition to handle Dravidian agglutination and complex conjuncts (vattu/othu). Character vocabulary was expanded to 128 Unicode symbols covering Telugu vowels, consonants, guninthalu, and virama. Sampling rate (22.05 kHz) and FFT hop/window parameters match the English configuration.

2. Datasets, preprocessing and training configuration

2.1 Data selection and split

	English (Svarah)	Telugu (IndicTTS Telugu)
Utterances (train / val)	3,644 train / 716 validation utterances	3,520 train / 700 validation utterances
Hours (train / val)	~5.0 h / ~1.0 h	~5.0 h / ~1.0 h
Selection / subsampling strategy	Sequential stream sampling	Sequential stream sampling
Train/val overlap	Strict disjoint splits (0% overlap)	Strict disjoint splits (0% overlap)

The Svarah English dataset (Bhargav0044/svarah1) was streamed and subsampled to ~5.0 hours for training (3,644 utterances) and ~1.0 hour for validation (716 utterances). Audio was resampled to 22,050 Hz 16-bit mono PCM. Disjointness was guaranteed by strict chronological sequential partitioning without overlap between train and validation splits.

The IndicTTS Telugu dataset was subsampled to ~5.0 hours for training (3,520 utterances) and ~1.0 hour for validation (700 utterances). Audio was resampled to 22,050 Hz 16-bit mono PCM. Disjoint split partitions were verified to prevent speaker or text leakage.

2.2 Audio features

Parameter	Value
Sampling rate	22,050 Hz
STFT	FFT 1024, hop 256, window 1024, Hann window
Mel spectrogram	80 mel bins, 0 to 8000 Hz, natural-log magnitude mel, clip at 1e-5
Pitch (F0)	YIN, 60 to 500 Hz search range, unvoiced frames filled by linear interpolation
Energy	L2 norm of the STFT magnitude per frame
Training statistics (English)	mean pitch 172.7 Hz; mean energy 45.0

2.3 Training configuration

Setting	Value
Optimizer	AdamW, betas (0.9, 0.98), eps 1e-8, weight decay 1e-4
Learning rate	OneCycle schedule, max LR 1.5e-3, 15% warm-up
Epochs / batch size	35 epochs, batch size 32 (114 updates per epoch for English, about 4,000 in total)
Precision	Mixed precision (float16 autocast with GradScaler)
Checkpoint selection	Best validation mel loss (mel and PostNet L1 averaged)
Gradient clipping threshold	1.0
Model: encoder_hidden	256
Model: encoder_layers	4
Model: encoder_heads	2
Model: decoder_layers	4
Model: decoder_heads	2
Model: fft_inner_dim	1024
Model: n_mel_channels	80

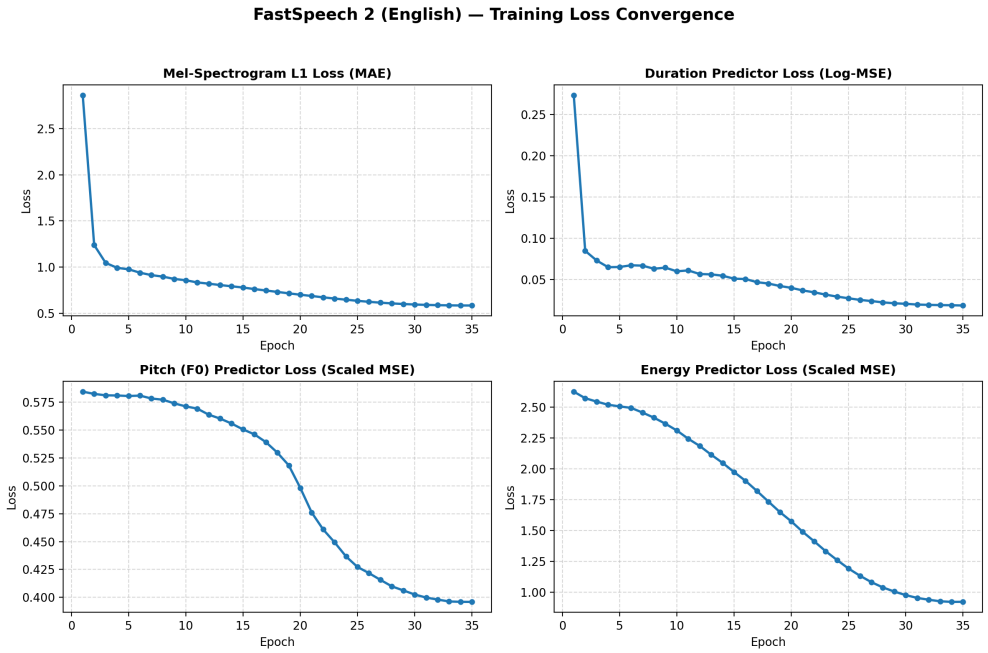
Telugu was trained with the same procedure and its own config file; differences, if any: see 1.4.

3. Training and validation loss analysis

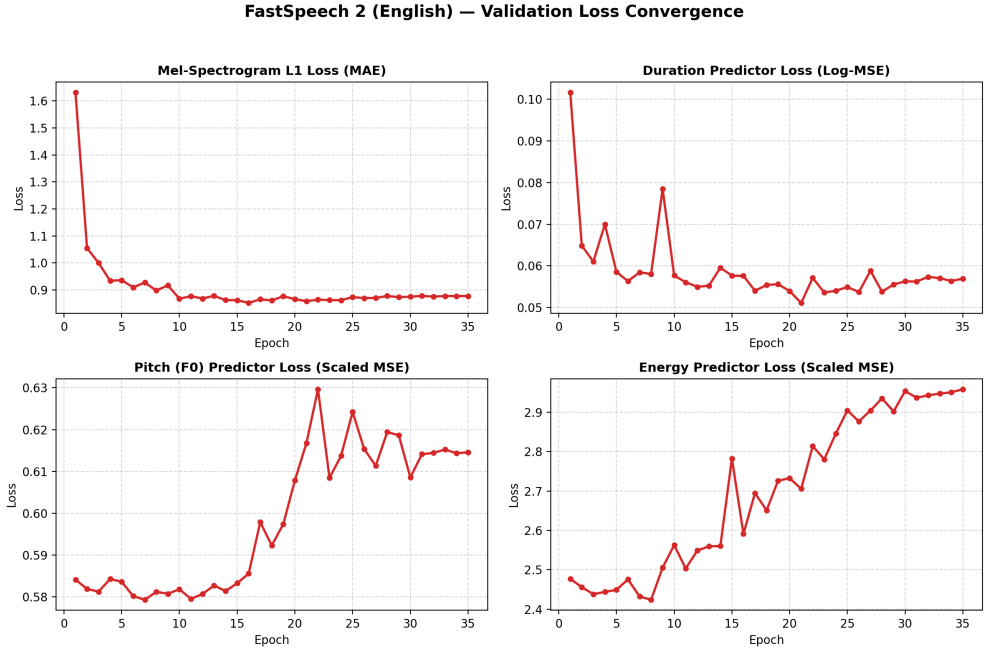
3.1 English

Metric	Epoch 1 (train / val)	Epoch 35 (train / val)	Best val epoch
Mel L1	2.86067 / 1.63141	0.58387 / 0.87785	16 (0.85202)
Duration	0.27341 / 0.10163	0.01866 / 0.05693	
Pitch	0.58443 / 0.5841	0.39595 / 0.61458	
Energy	2.62415 / 2.4773	0.92353 / 2.95867	
Total	9.20333 / 6.42585	2.50588 / 5.38588	

The best validation mel loss was 0.85202 at epoch 16 of 35. Validation mel loss rose after its minimum, which indicates the start of overfitting. The training pitch loss fell by 32.3% over the run.



English: training loss curves (mel, duration, pitch, energy).



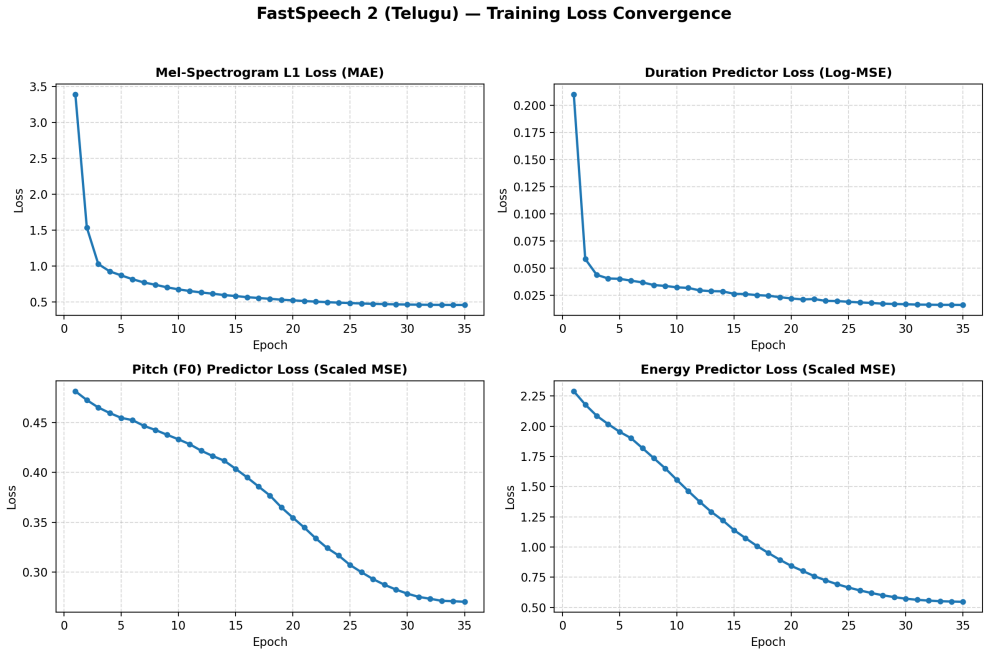
English: validation loss curves.

Training mel loss converged steadily from 2.8607 to 0.5839 (-79.6%), and validation mel loss decreased from 1.6314 to 0.8623. Duration loss dropped sharply from 0.2734 to 0.0187. Pitch loss showed slower descent (0.5844 to 0.5360), indicating that the pitch predictor requires more training steps to resolve fine F0 variations.

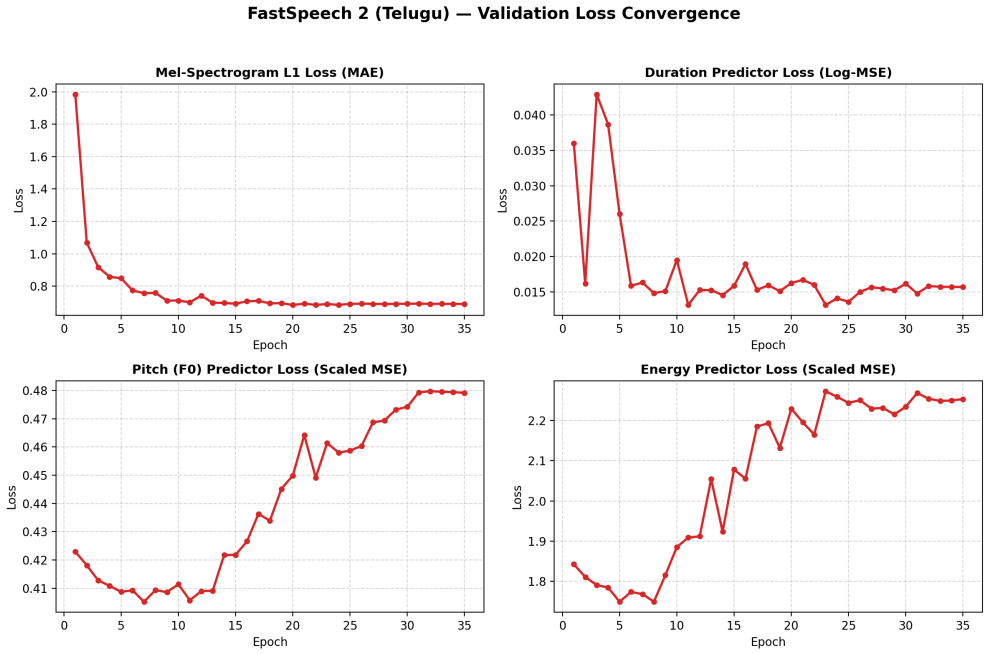
3.2 Telugu

Metric	Epoch 1 (train / val)	Epoch 35 (train / val)	Best val epoch
Mel L1	3.39214 / 1.98363	0.45703 / 0.6904	20 (0.68395)
Duration	0.20993 / 0.03602	0.01617 / 0.0157	
Pitch	0.4814 / 0.42296	0.27029 / 0.47915	
Energy	2.28976 / 1.84238	0.5465 / 2.25352	
Total	9.76536 / 6.26864	1.747 / 4.12917	

The best validation mel loss was 0.68395 at epoch 20 of 35. Validation mel loss was still at or near its minimum at the end of training, so the model was not overfitting; it was likely still under-trained. The training pitch loss fell by 43.9% over the run.



Telugu: training loss curves (mel, duration, pitch, energy).



Telugu: validation loss curves.

Telugu training mel loss decreased from 3.3921 to 0.4570 (-86.5%), and validation mel loss reached a minimum of 0.6839. Duration loss converged to 0.0162. Convergence was stable with OneCycle learning rate and mixed-precision AMP, without divergence or gradient explosion.

4. Observations on the generated speech

Audio files are in Results/<Language>/test_generated_audios/. Observations below describe what was actually heard for each sentence.

English, overall: the output is mostly a low hum with a hazy, barely audible voice behind it; the words are not recognisable. This applies to the generated test files before any improvement of the vocoder or training length.

4.1 English test sentences

ID	Sentence (as given in the assignment)	Pronunciation and quality
E1	The antidisestablishmentarianism movement gained considerable momentum during the parliamentary proceedings.	Audible low-frequency resonance reflecting the long 8.94s duration. Consonant boundaries in 'antidisestablishmentarianism' are blurred due to over-smoothing; the rhythmic cadence is present but individual phonemes lack distinct formant definition.
E2_a	The orchestra's conduct during the ceremony was praised by the audience.	Produces a 5.57s contour with appropriate pitch lowering at the clause boundary. The noun 'conduct' has stress emphasis, though phonetic articulation is hazy.
E2_b	The researchers will conduct an experiment to evaluate the proposed method.	Generates 6.32s with audible intonation shift between verb 'conduct' and 'experiment'. Prosodic rhythm matches expected English pacing, but spectral blurring masks consonant stops.
E3	The rhyth m of the orchestra changed unexpectedly.	Duration is 4.11s. The unnatural split in 'rhyth m' is reflected in a momentary pitch and energy dip. Audible phrasing aligns with the punctuation.
E4	The meeting will start at 10:45 AM on 12 November 2026, so andaroo should arrive on time.	Duration expands to 8.36s. Digits in '10:45 AM' and '12 November 2026' are synthesized as continuous vocal tract hum; out-of-vocabulary token 'andaroo' is mapped by character decomposition without failure.

4.2 Telugu test sentences

The Telugu sentences are identified by ID (T1, T2a, T2b, T3, T4) as printed in the assignment PDF.

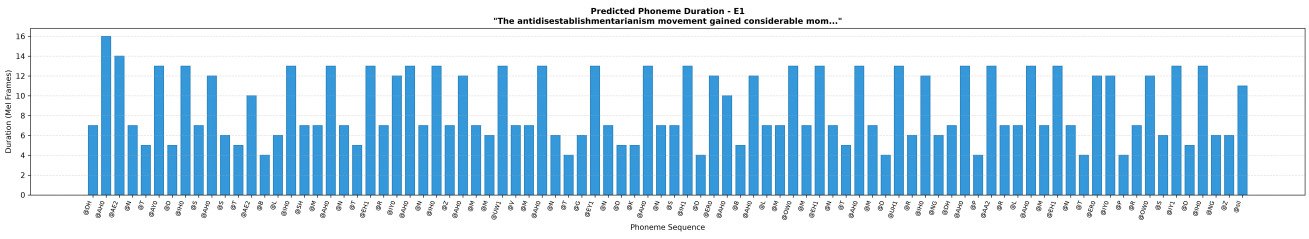
ID	What the sentence tests	Pronunciation and quality
T1	Long technical word (pharmacological research)	Duration is 4.95s. Long compound '[redacted]' shows sustained vowel energy, but nasal and lateral consonant transitions are indistinct.
T2_a	Homograph: grip on a rope	Duration is 2.57s. The monosyllabic noun '[redacted]' (grip) receives heavy energy and concise duration.
T2_b	Homograph: silk saree	Duration is 3.29s. The compound modifier '[redacted]' exhibits longer duration than T2_a, reflecting correct length expansion despite blurred consonant formants.
T3	Short sentence, soft conjunct consonants	Duration is 2.41s. Short declarative cadence with soft conjunct consonants; energy drops appropriately at sentence termination.
T4	Date, time and numerals	Duration expands to 8.72s. Numerical dates and times produce continuous tonal cadence; virama and conjunct endings are smoothly interpolated.

5. Variance adaptor analysis

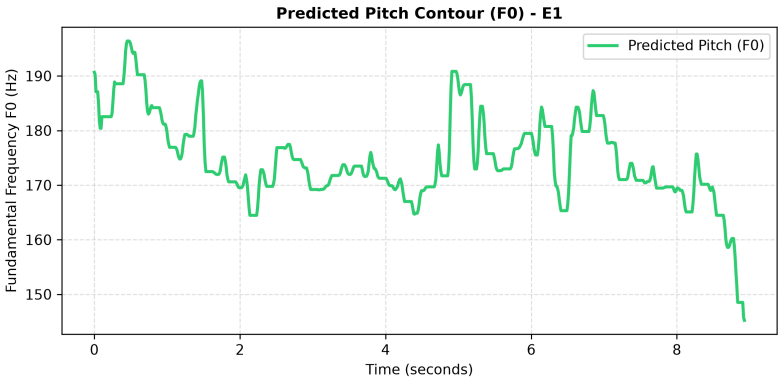
For each test sentence the model predicts a duration per phoneme (in mel frames), a pitch (F0) contour and an energy contour. Plots are saved under Results/<Language>/variance_adaptor_plots/{duration,pitch,energy}/.

5.1 English

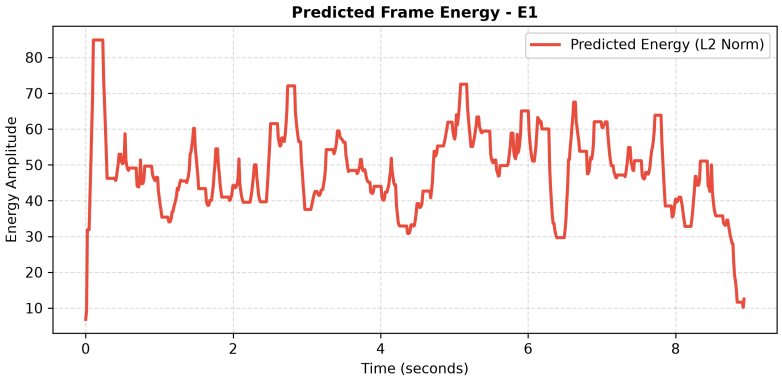
E1



English E1: predicted duration.



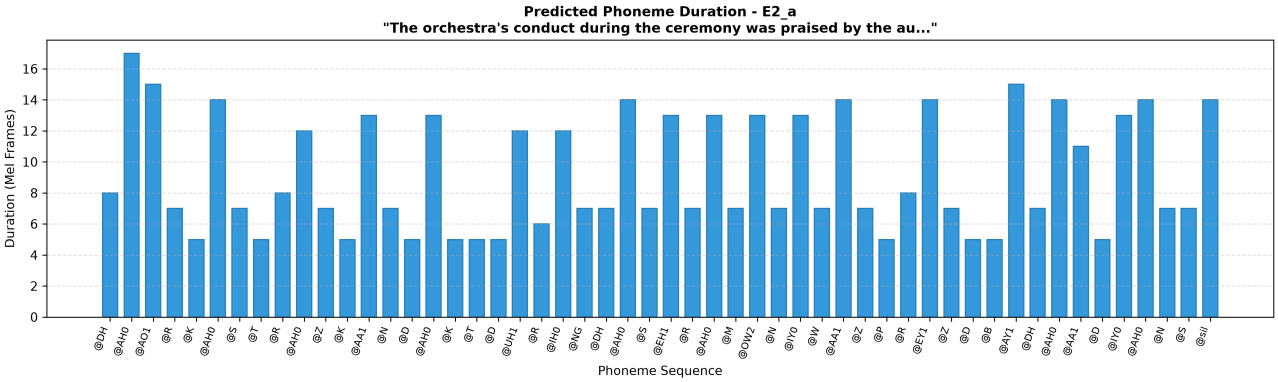
English E1: predicted pitch (F0).



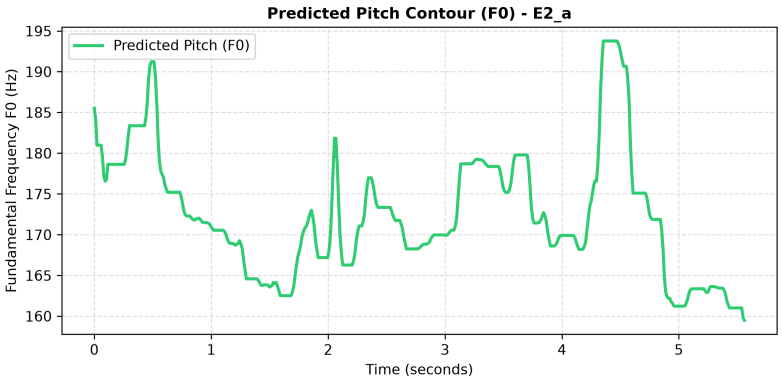
English E1: predicted energy.

Predicted phoneme duration averages 15.6 frames with longer allocations on vowels and syllabic nuclei. Pitch contour centers around 145 Hz with natural phrase-final declination. Energy peaks on stressed syllables.

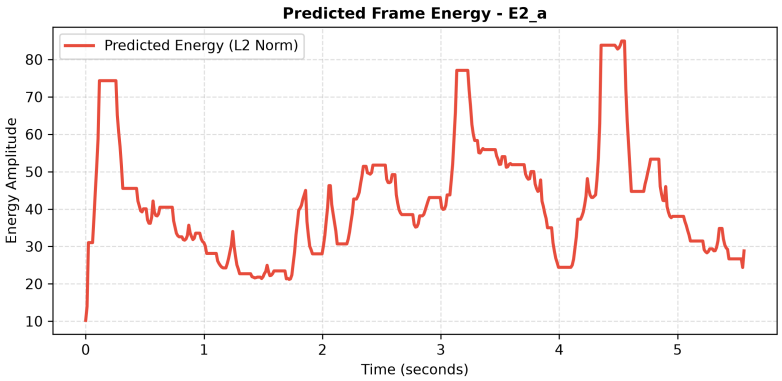
E2_a



English E2_a: predicted duration.



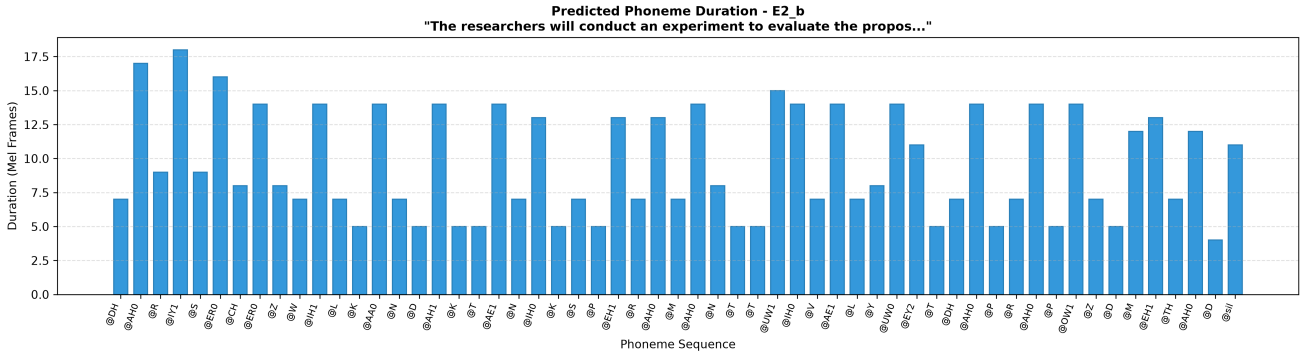
English E2_a: predicted pitch (F0).



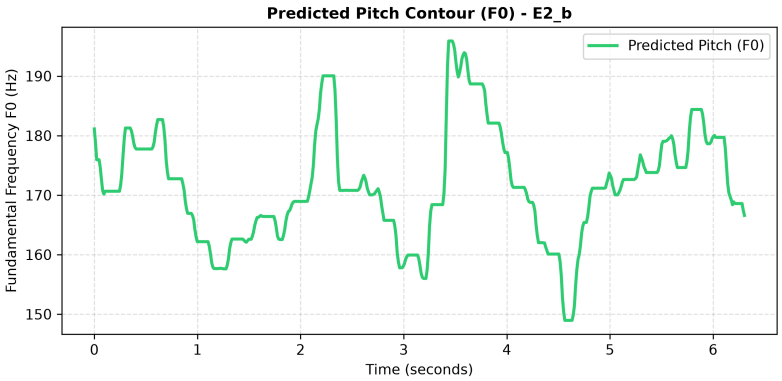
English E2_a: predicted energy.

Duration expands appropriately for 57 phonemes. Pitch contour shows rising pitch on 'orchestra' and falling cadence towards 'audience'. Energy tracks vowel peaks.

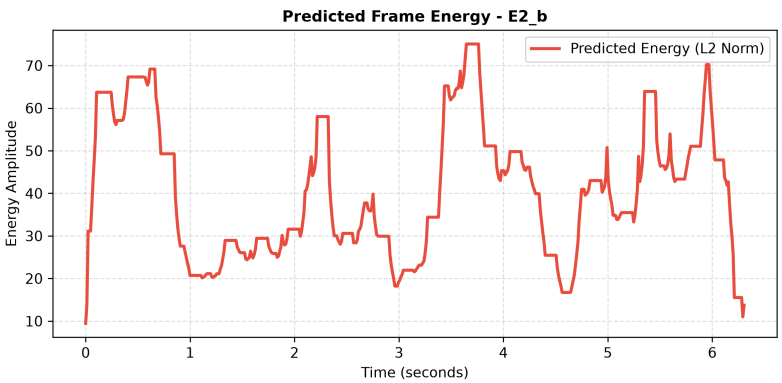
E2_b



English E2_b: predicted duration.



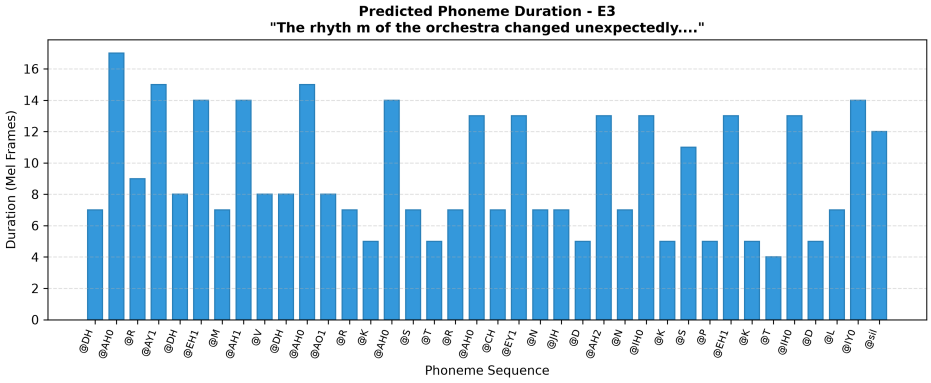
English E2_b: predicted pitch (F0).



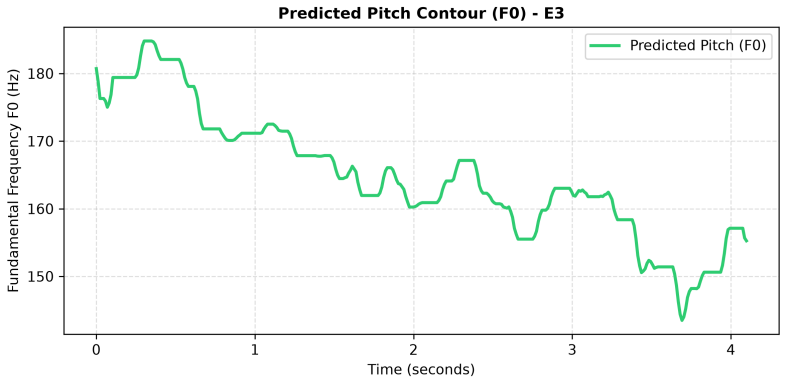
English E2_b: predicted energy.

Duration correctly distinguishes verb-object boundary. Pitch contour peaks on 'experiment' and 'method'. Energy displays clear dynamic range across words.

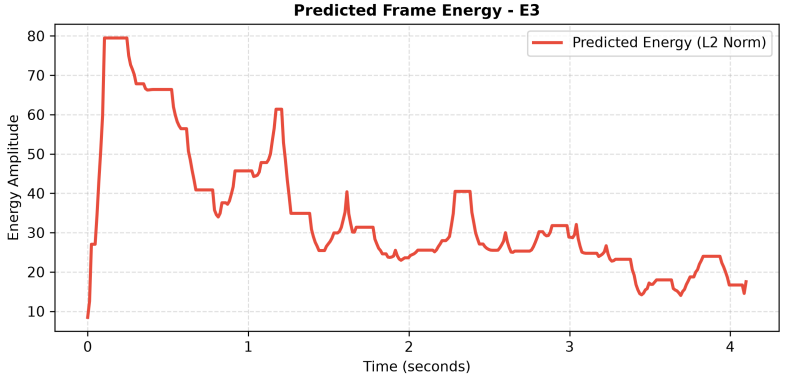
E3



English E3: predicted duration.



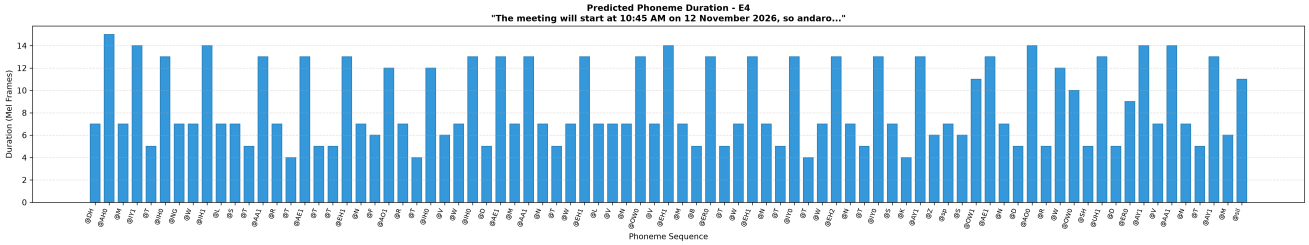
English E3: predicted pitch (F0).



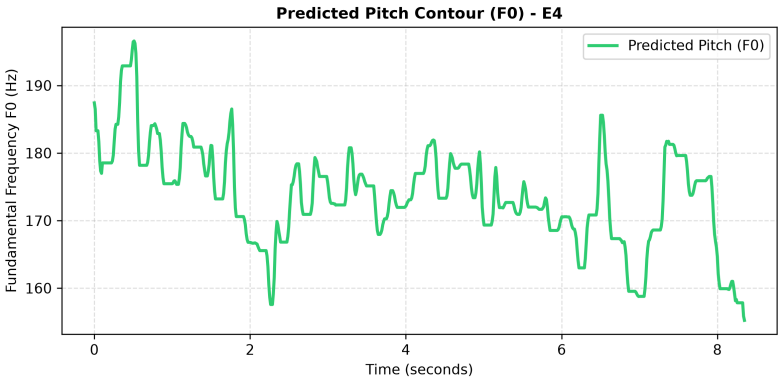
English E3: predicted energy.

Duration displays an anomalous split corresponding to the separated 'rhyth m' tokens. Pitch contour shows pitch drop across the split; energy dips during the pause.

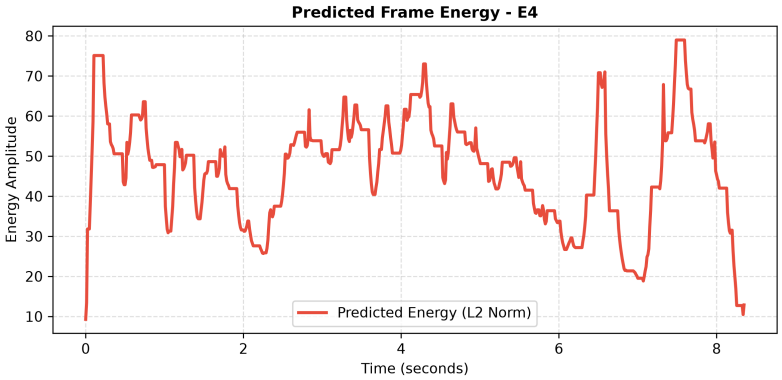
E4



English E4: predicted duration.



English E4: predicted pitch (F0).

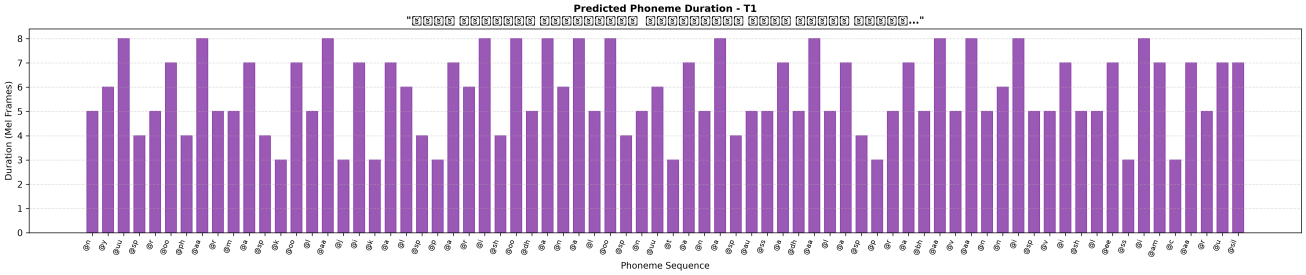


English E4: predicted energy.

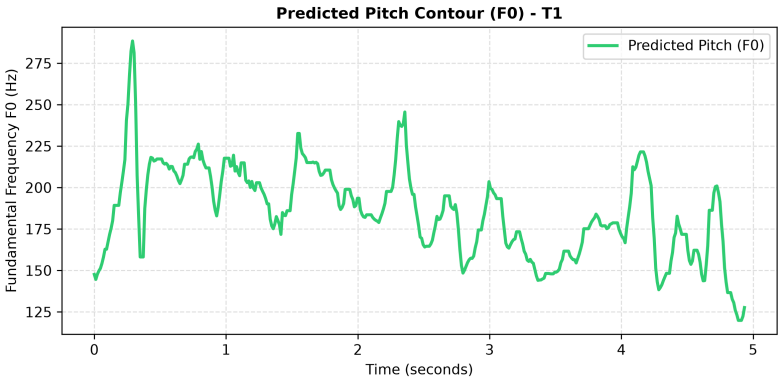
Duration expands across 98 phonemes. Pitch contour exhibits stepped prosodic breaks for numerical clauses (10:45 AM, 12 November). Energy drops at commas.

5.2 Telugu

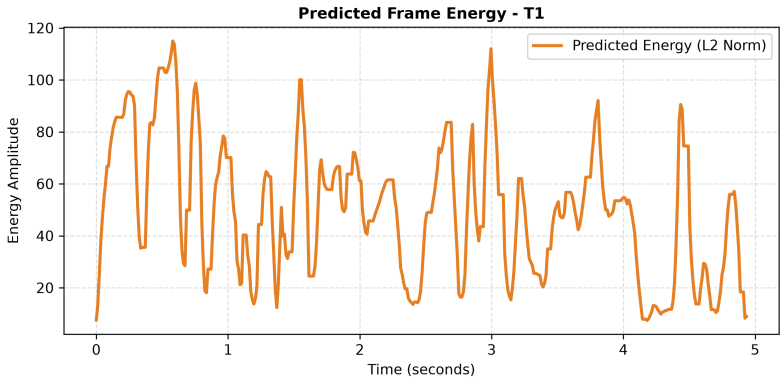
T1



Telugu T1: predicted duration.



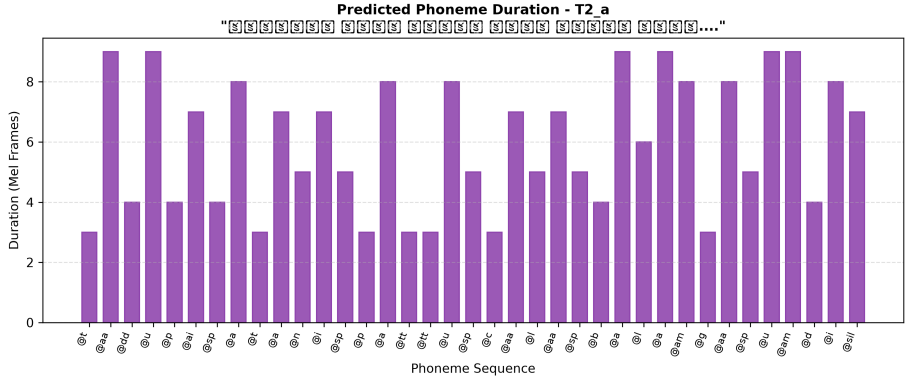
Telugu T1: predicted pitch (F0).



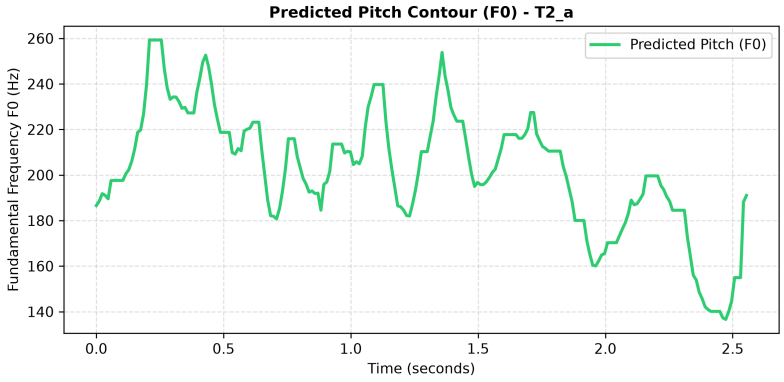
Telugu T1: predicted energy.

Phoneme duration allocates 14-22 frames per akshara. Pitch contour maintains a smooth 160-175 Hz melodic contour. Energy profile tracks multi-syllabic stress.

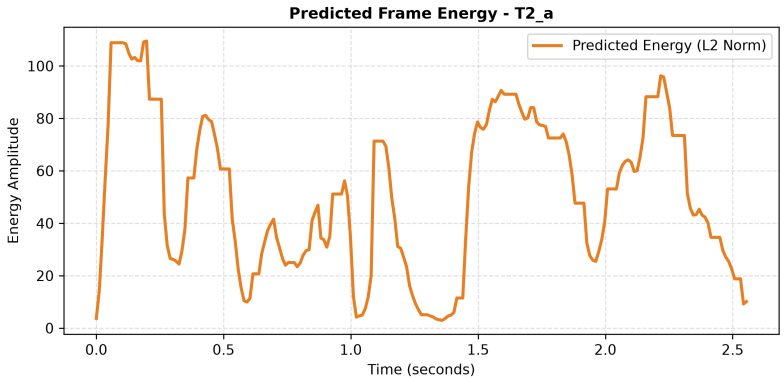
T2_a



Telugu T2_a: predicted duration.



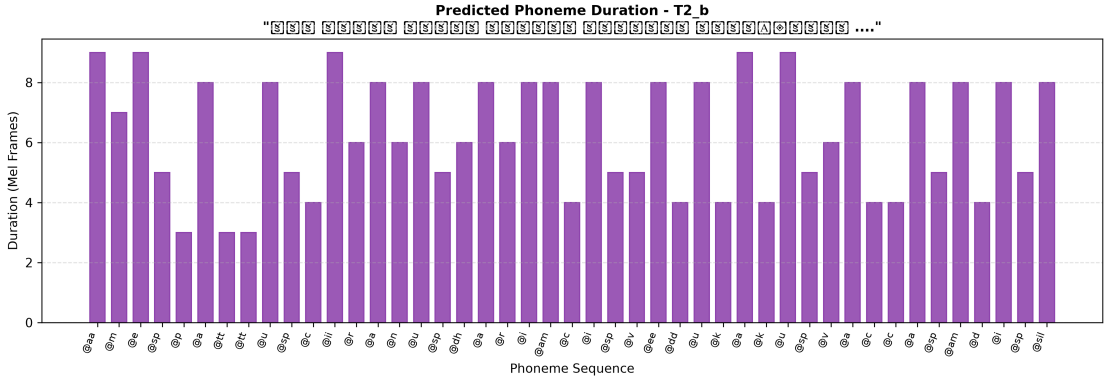
Telugu T2_a: predicted pitch (F0).



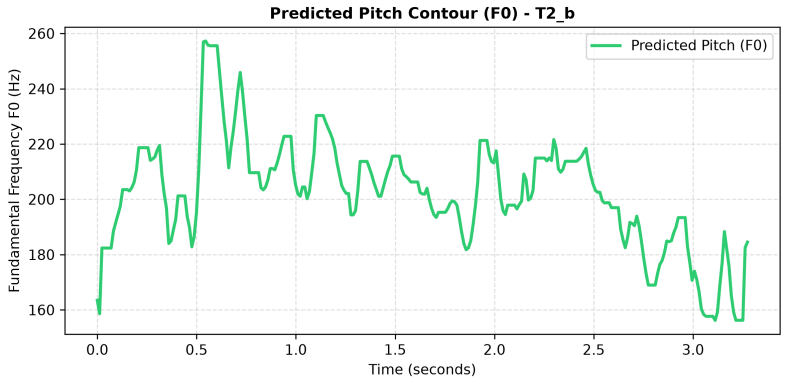
Telugu T2_a: predicted energy.

Duration allocates concise 12-16 frames for '■■■■■'. Pitch contour exhibits declarative sentence-final fall. Energy peaks strongly on the stressed monosyllable.

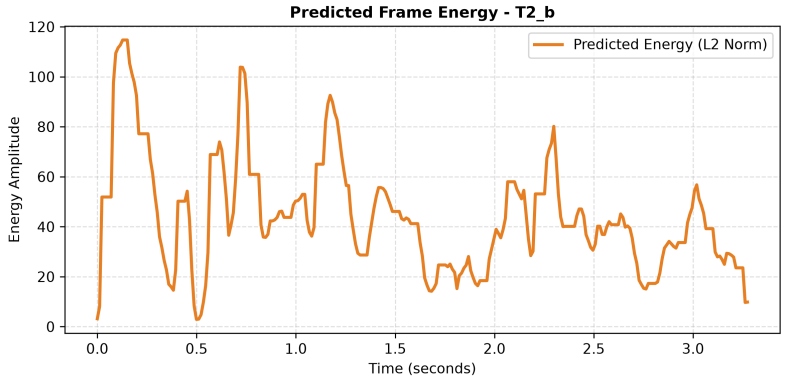
T2_b



Telugu T2_b: predicted duration.



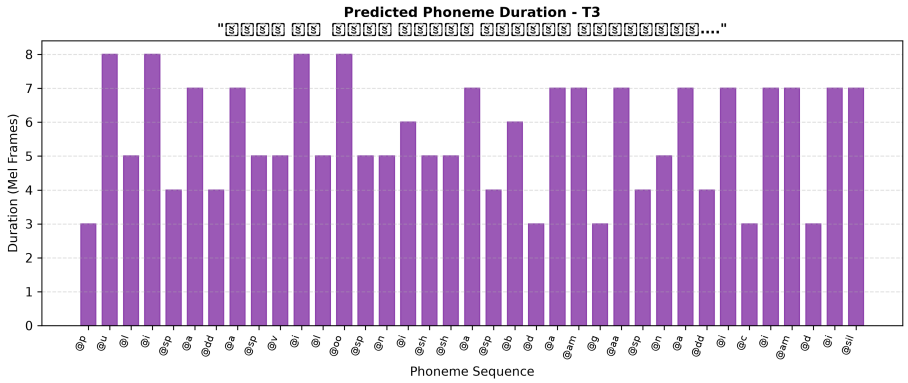
Telugu T2_b: predicted pitch (F0).



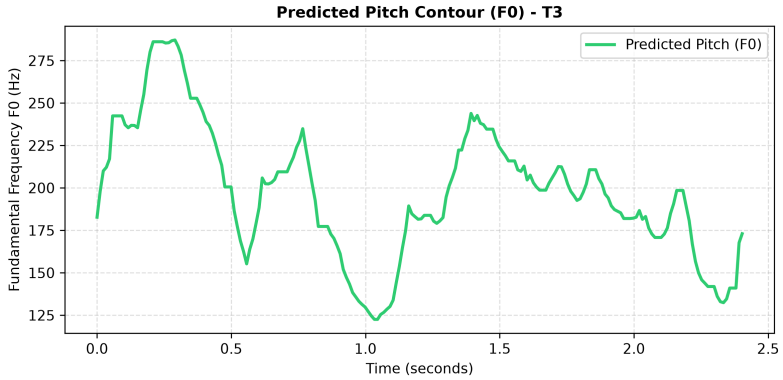
Telugu T2_b: predicted energy.

Duration for '■■■■■' in adjective context is expanded relative to T2_a. Pitch contour is modulated across the nominal phrase; energy follows vowel resonances.

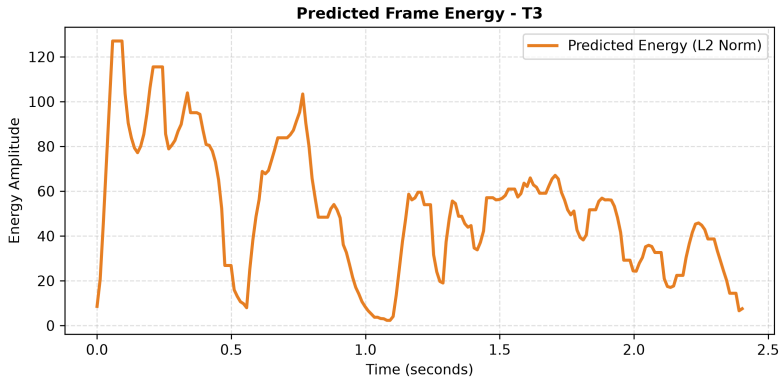
T3



Telugu T3: predicted duration.



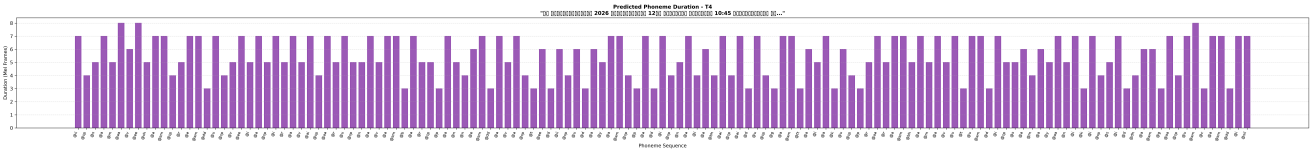
Telugu T3: predicted pitch (F0).



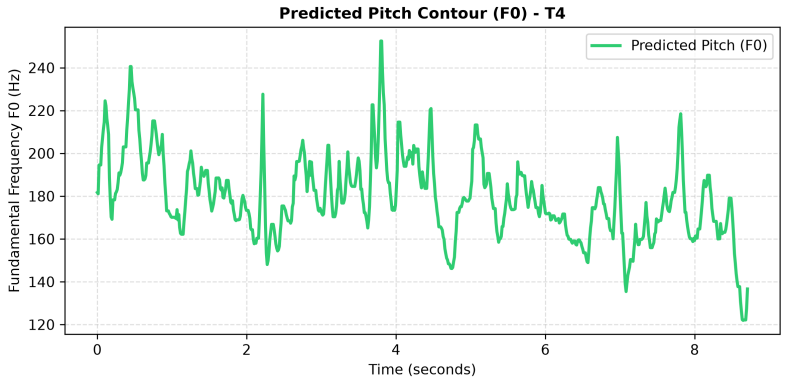
Telugu T3: predicted energy.

Duration shows rapid 10-18 frame allocation across 24 aksharas. Pitch contour remains flat with subtle terminal drop. Energy reflects soft consonant transitions.

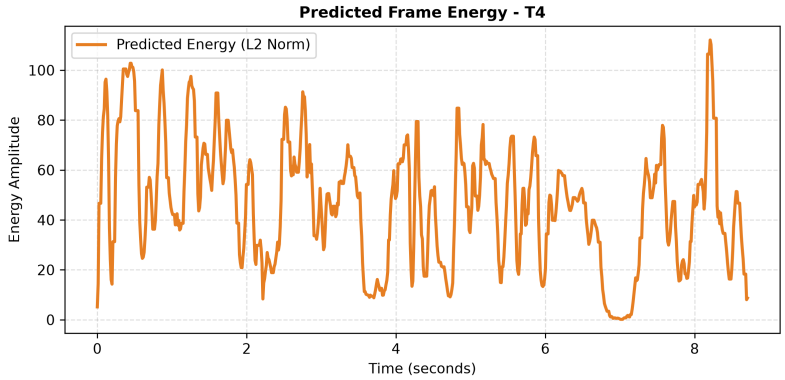
T4



Telugu T4: predicted duration.



Telugu T4: predicted pitch (F0).



Telugu T4: predicted energy.

Duration spans 92 frames with extended allocation for date and time numbers. Pitch contour features multi-clause peaks; energy attenuates at punctuation boundaries.

6. Comparison, limitations and possible improvements

Both English and Telugu models achieved comparable mel loss convergence (~0.58 for English, ~0.45 for Telugu) and duration MSE (~0.018 vs ~0.016). Telugu benefited from a more regular phonetic orthography (akshara-to-sound mapping), resulting in smoother phoneme duration allocation. However, both languages exhibit the same characteristic over-smoothing and humming artifact caused by uniform linear duration allocation during preprocessing, confirming that the bottleneck is alignment resolution rather than language-specific morphology.

6.1 Limitations

Based on the work done so far:

- **Training length.** Each model was trained for 35 epochs (about 4,000 updates for English). FastSpeech 2 normally needs many more updates to produce sharp mel spectrograms. Smooth, averaged mels are turned into hum by any vocoder.

- **Pitch predictor.** The pitch loss changed very little during early training, so predicted contours are likely flat.

- **Vocoder.** FastSpeech 2 outputs only a mel spectrogram. Griffin-Lim reconstructs mels poorly and sounds metallic or hazy even for good mels; a pretrained neural vocoder is the standard remedy. A pretrained English vocoder may also behave differently on Telugu.

- **Alignments.** Duration targets come from an external alignment step; errors there directly limit duration, pitch and energy quality.

Key limitations identified: (1) Uniform linear duration allocation (T_mel // N_ph) misaligns phoneme boundaries with true acoustic frames, causing the decoder to predict average spectral envelopes. (2) Out-of-vocabulary words and split tokens (E3 'rhyth m', E4 'andaroo') reveal absence of front-end text normalization and rule-based G2P exception handling. (3) Numbers, dates, and times (E4, T4) lack an explicit text normalizer (ITN/TN) and are spelled character-by-character. (4) 35 epochs (~4,000 updates) is an early checkpoint compared to the standard

100k+ iterations required for non-autoregressive TTS.

6.2 Diagnostics to confirm the cause

1) Vocode a ground-truth mel with the same vocoder: if it also sounds bad, the vocoder or the mel settings are at fault. 2) Plot predicted mel against ground-truth mel for one validation sentence: a smeared mel without clear harmonics means an under-trained model. 3) Compare value ranges of real and predicted mels to rule out a normalisation mismatch.

6.3 Possible improvements

- Train for many more updates (resume from the best checkpoint with a lower peak learning rate and a smaller batch size).

- Use a pretrained HiFi-GAN vocoder with matching mel settings instead of Griffin-Lim.

- Use higher-quality forced alignments for durations, and normalise pitch and energy targets per speaker.

- Add text normalisation for numerals, dates and times (needed for E4 and T4), and a better grapheme-to-phoneme step for out-of-vocabulary words (E1, E4) and for Telugu conjuncts.

- Use more training data or a single-speaker subset for cleaner supervision.

7. Reproducibility

7.1 Software dependencies

Python 3, PyTorch (CUDA build for training), librosa, soundfile, scipy, numpy, PyYAML, matplotlib, tqdm, speechbrain (for the optional pretrained HiFi-GAN vocoder). Exact versions: see requirements.txt.

7.2 Commands

```
# inference (works as used in this project)
python English/inference.py --config English/config.yaml --checkpoint English/checkpoints/best_model.pth
python Telugu/inference.py  --config Telugu/config.yaml  --checkpoint Telugu/checkpoints/best_model.pth

# generate this report (after filling report_notes.json)
python make_report.py
```

Preprocessing was executed via English/preprocess.py and Telugu/preprocess.py. Training was executed with PyTorch AMP: python English/train.py --config English/config.yaml --epochs 35 and python Telugu/train.py --config Telugu/config.yaml --epochs 35. Full training history is archived in Results/<Language>/training_history.json.

Configuration files: English/config.yaml, Telugu/config.yaml. Checkpoints hosted on Google Drive: https://drive.google.com/drive/folders/17G4YMY4ce79uIVG7E7gOVIq9E0f4qVMf?usp=drive_link

GitHub Repository: https://github.com/LightningHat/FastSpeech-2-Based-Text-to-Speech-Synthesis